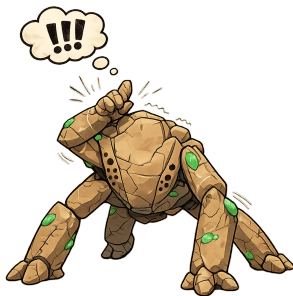


PHS Hail Mary Orientation

Correlation and Linear Regression



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Today, we'll be building on Dylan, Tillie, and Ashlynn's lessons on math, probability, variance, and covariance by discussing correlation and linear regression.

I'm hoping the lessons are feeling well-paced and accessible, but PLEASE interrupt me if I'm not making sense or (especially) you disagree with what I'm saying!

Key skills and concepts

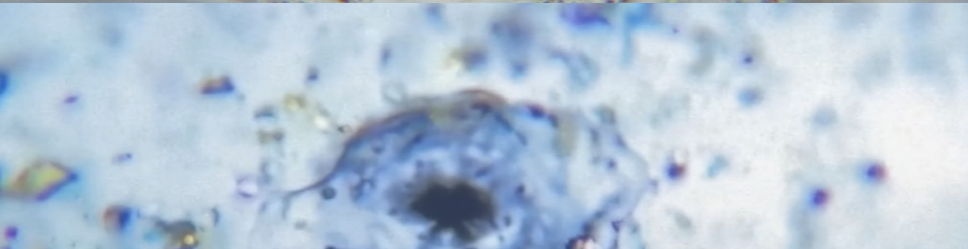
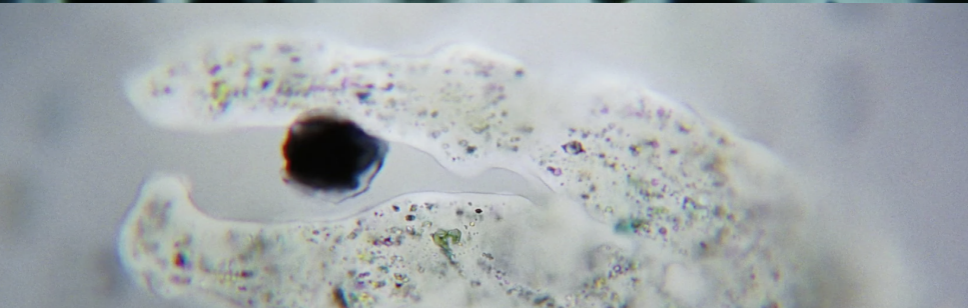
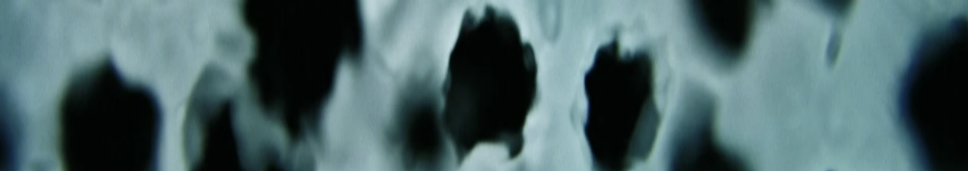
- Calculating Pearson's r
- Enumerating key features and drawbacks of Pearson's r
- Reading linear regression model notation
- Describing linear regression estimates in plain language

Correlation

Motivating example

Astrophage are microbial star-eaters which threaten death upon any solar system they infest.

- Grace and Rocky have discovered an Astrophage-eating predator endogenous to the Tau Ceti solar system. They name it, with great inspiration, Taumoeba.
- The intrepid duo watch Taumoeba eat Astrophage and reason that since Tau Ceti is the only star in their cluster without mass-extinction level dimming, Taumoeba must prevent Astrophage-induced star death.



Motivating example

...How do we know there is truly some kind of relationship between these two organisms, apart from the single observation made by Grace?

...What statistical tools can we use to explore the relation between Taumoeba presence in a solar system and Astrophage population density?

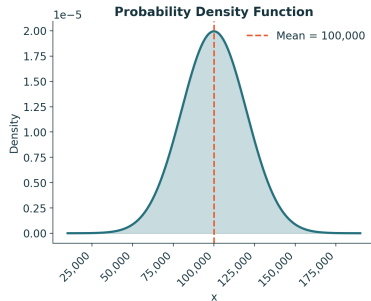
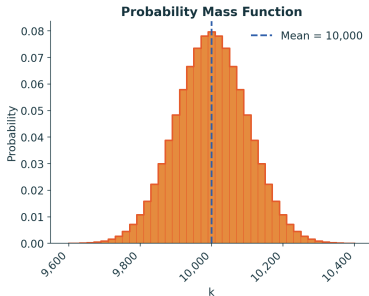
...These are questions to which Earth needs answers!!!!!!

...Looks like a job for Harvard's 2026
PHS PhD G1's 

Quantifying the problem

- X = number of Taumoeba in a solar system*
- Y = number of Astrophage in a solar system*
- $\bar{X}$ = average number of Taumoeba per solar system
- $\bar{Y}$ = average number of Astrophage per solar system

* What type of random variables are X and Y , discrete or continuous?



Approaching the problem with correlation

Pearson's Sample Correlation

$$r_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

The three components of this formula have names.

Can anyone tell me what they are?

Approaching the problem with correlation

Pearson's Sample Correlation

$$\rho_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

- Covariance (numerator)
- Standard deviation of x (denominator)
- Standard deviation of y (denominator)

Correlation versus covariance

We see in this formula slide that the correlation between two random variables is just the covariance of those variables standardized into a unitless quantity.

What advantages might this have?

Correlation versus covariance

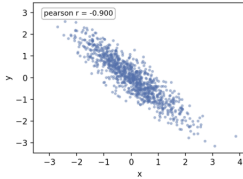
Correlation advantages

1. Scale free, resistant to unit changes
2. Comparability
3. Bounded range $[-1, 1]$
4. Consistent relationship between strength and size

Visualizing Pearson's correlation

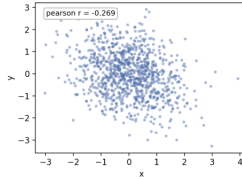
Strong Negative

target $r = -0.9$



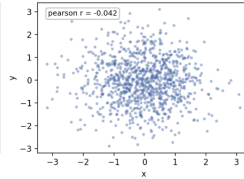
Weak Negative

target $r = -0.3$



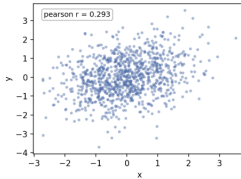
Null

no relationship



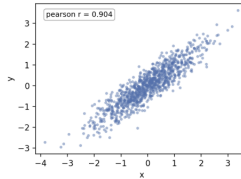
Weak Positive

target $r = 0.3$



Strong Positive

target $r = 0.9$



Completing the problem

Table 1: Local stellar neighborhood Taumoeba and Astrophage populations

Solar System	Taumoeba	Astrophage
Sol	1	1,000,000
40 Eridani	2	1,500,000
Epsilon Eridani	1,400	800,000
Sirius	7,000	700,000
WISE 0855-0714	2,000	900,000
Tau Ceti	400,000	1,500
-	$\bar{x} = 68,400.5$	$\bar{y} = 816,916.67$

Completing the problem

Table 2: Applying components of the correlation formula to Table 1

System	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$	$(x_i - \bar{x})(y_i - \bar{y})$
Sol	-68,399.5	+183,083.3	4.68e9	3.35e10	-1.25e10
40 Eridani	-68,398.5	+683,083.3	4.68e9	4.67e11	-4.67e10
Epsilon Eridani	-67,000.5	-16,916.7	4.49e9	2.86e8	+1.13e9
Sirius	-61,400.5	-116,916.7	3.77e9	1.37e10	+7.18e9
WISE 0855-0714	-66,400.5	+83,083.3	4.41e9	6.90e9	-5.52e9
Tau Ceti	+331,599.5	-815,416.7	1.10e11	6.65e11	-2.70e11
		Sum	1.320e11	1.186e12	-3.268e11

Completing the problem

$$r_{xy} = \frac{-3.268 \times 10^{11}}{\sqrt{1.320 \times 10^{11}} \times \sqrt{1.186 \times 10^{12}}} = \frac{-3.268 \times 10^{11}}{363,295 \times 1,088,984}$$

Result

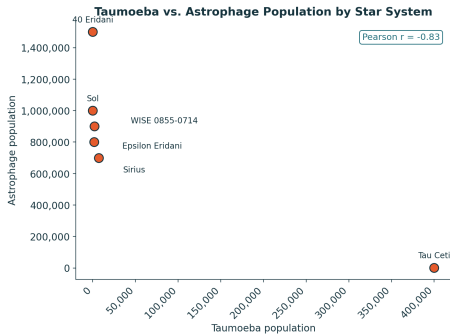
$$r_{xy} \approx -0.826$$

Assumptions of Pearson's Sample Correlation

Assumptions

1. Continuous scale
2. Linear relationship
3. No outliers
4. Normality
5. Independence of observations

Can you think of violations within this example?



1. Which is stronger, a correlation of -0.8 or of 0.8 ?
2. What is the correlation of a random variable with itself?
3. What is the correlation between two independent random variables?
4. If two variables have covariance and correlation equal to zero, are they always independent?

Answers

1. They're equal! Magnitude, not sign, determines correlation strength.
2. r_{xx} always equals 1!
3. 0! Independence guarantees the correlation numerator equals 0.
4. No! Correlation captures linear relationships only.

Correlation $\neq$ causation!

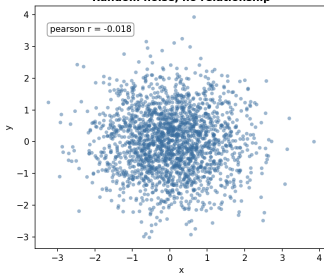
Causation is just one possible explanation for the r_{xy} we see, and in our real-world research, an unlikely one at that!

But...

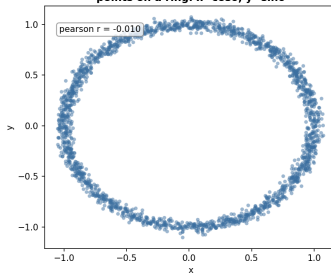
Causation also does not always imply correlation!

Non-linear dependence and Pearson's r

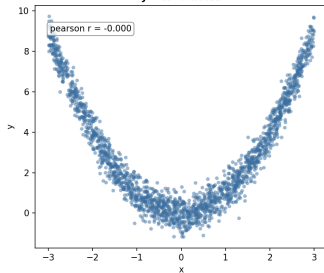
True Independence
Random noise, no relationship



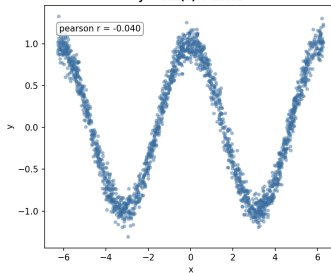
Circular Dependence
points on a ring: $x = \cos\theta$, $y = \sin\theta$



Quadratic Dependence
 $y = x^2 + \text{noise}$



Cosine Dependence
 $y = \cos(x) + \text{noise}$



Questions?

Linear Regression

Beyond Correlation

- An important question we definitely want to answer: For each additional Taumoeba, how many additional/fewer Astrophage do we observe?
 - NOTE: this question does not ask about the causal effect of Taumoeba population density on Astrophage population density.
- Correlation and covariance do **not** provide an answer to such questions!
- Linear regression extends beyond correlation to quantify how much one variable changes with each additional increase in another variable.
- Linear regression models can consider these questions **AND** can be extended to include **multiple relevant covariates!**

Linear Regression

To begin, we can make the modeling assumption that the relation between X and Y can be drawn using a straight line.

- $y = mx + b \leftrightarrow y = b + mx$

Mathematical notation allows us to express complex ideas precisely.

- $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$
- $E[Y|X = x] = \beta_0 + \beta_1 x$

b, β_0 are intercept parameters. m, β_1 are slope parameters.



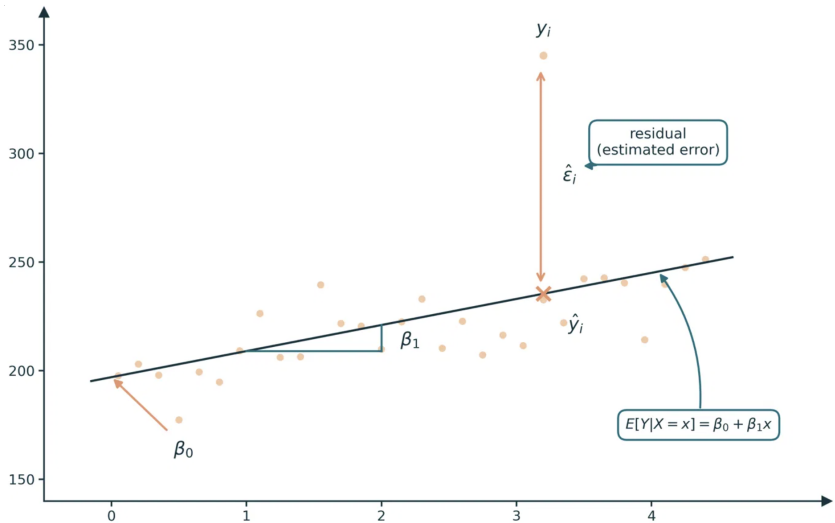
LabRule

@lab_rule

$Y = mx + b$ jokes are great but at
some point we gotta draw the line

8:22 AM · 03 Mar 21 · [Twitter for Android](#)

Visualizing regression parameters



Recall the notation lesson...

1. What is the difference between β_0 and $\hat{\beta}_0$?
2. What is the difference between β_1 and $\hat{\beta}_1$?
3. When do we see β versus $\hat{\beta}$?

Answers!

1. β_0 is a population parameter and $\hat{\beta}_0$ is a sample statistic.
2. The difference is the same. The hat notation applies across all beta coefficients.
3. We almost always calculate $\hat{\beta}$ because we work with samples. The true population parameter is only visible when we work with data from an entire population (e.g., faculty salary data from all accredited public health schools in the US.)

Estimators and Estimation

Parameters from linear models are frequently estimated using ordinary least squares (OLS) or maximum likelihood estimation (MLE) estimators. We'll get to this in future lessons this semester!

- The OLS solution for the best $\hat{\beta}_1$ estimate of the β_1 parameter given observed data is:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Does this look familiar??

How the $\hat{\beta}_1$ formula relates to correlation

1) Correlation formula:

$$r = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

Step 2: Multiply by $\sqrt{\text{Var}(Y)}$

2) Multiply by $\sqrt{\text{Var}(Y)}$:

$$r \cdot \sqrt{\text{Var}(Y)} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \cdot \cancel{\sqrt{\text{Var}(Y)}}} \cdot \cancel{\sqrt{\text{Var}(Y)}}$$

Step 3: Divide by $\sqrt{\text{Var}(X)}$

3) Divide by $\sqrt{\text{Var}(X)}$:

$$r \cdot \frac{\sqrt{\text{Var}(Y)}}{\sqrt{\text{Var}(X)}} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}} / \sqrt{\text{Var}(X)}$$

4) Reaching the $\hat{\beta}_1$ formula from the previous slide:

$$r \cdot \underbrace{\frac{sd_Y}{sd_X}}_{\text{sample standard deviation}} = \boxed{\frac{\text{Cov}(X, Y)}{\text{Var}(X)}} = \frac{\text{covariance}(X, Y)}{\text{variance}(X)}$$

Questions?

Acknowledgements

Acknowledgments

HUGE thank you to the previous Teaching Fellows who contributed to these slides!

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- Matt Lee
- Louisa Smith
- Chris Boyer

And... to the creator of the metropolis beamer theme: Matthias Vogelgesang!